

Marine A. Denolle

Associate Professor, Earth and Space Sciences, University of Washington | College of Environment CV — reverse chronological order; active/completed 2023–present appear first

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🔖 Google Scholar 📺 scoped6259 📺 uwgeophysics6888 🏠 denolle-lab.github.io

Education

- 2008 – 2014 **Stanford University** Geophysics
 - Supervisor: Dr. Gregory Beroza
 - Co-supervisors: Dr. Eric Dunham, Dr. Jesse Lawrence
- 2007 – 2008 **Ecole Normale Supérieure - IGP** Geophysics
 - Supervisors: Dr. Satish Singh (IGP), Dr. David Bercovici (Yale)
- 2006 **Ecole Normale Supérieure** Earth Sciences
- 2004 – 2005 **Lycée Chateaubriand, Rennes** Physics-Mathematics

Research Impact Summary

Research Citations: 3,071 total (h-index: 24, i10: 48) –54.4% from 2024 to 2025 (OpenAlex) | 3,513 total, h-index: 25 (Google Scholar, last known)

Publications: 68 peer-reviewed | 6 Nature/Science-family (T1) | 40 AGU-flagship (T2) | 21 domain journals (T3) | 1 preprints

Research Funding: Lead PI: \$3.1M (14 grants) | Co-PI/Co-I: \$4.9M (8 grants) | Fellowships: \$878K (2)

Open-Source Software: 412 GitHub stars | 267 forks | 20 active repos | NoisePy: 211 stars, 83 forks, 19 contributors

Mentoring: 8 PhD (4 current, 4 graduated) | 11 postdocs | 16+ undergrads | 10 co-supervised grad students | 5 became faculty

Broader Impact: mlgeo-book: 13 GitHub stars

Employment

- 2026 – present **Consultant**, Aligned HQ – Remote
Scientific Research & Frontier AI
- 2025 – present **Co-Founder**, Applied Environmental Intelligence – Seattle
- 2024 – present **Associate Professor**, University of Washington – Seattle, WA
Department of Earth and Space Sciences
- 2021 – 2024 **Assistant Professor**, University of Washington – Seattle, WA
Department of Earth and Space Sciences
- 2016 – 2021 **Assistant Professor**, Harvard University – Cambridge, MA
Department of Earth and Planetary Sciences
- 2014 – 2016 **Green Postdoctoral Fellow**, UC San Diego, Scripps Institution of Oceanography – La Jolla, CA
Institute of Geophysics and Planetary Physics
 - Supervisor Dr. Peter Shearer

Awards

- 2023 **Invited Professorship** Ecole Normale Supérieure rue d'Ulm-Paris
- 2023 **Data Science Fellow** eScience Institute, University of Washington
- 2019 **Charles F. Richter Early Career Award** Seismological Society of America
- 2019 **Kavli Frontiers of Science Fellow** National Academy of Sciences
- 2019 **Radcliffe Assistant Professorship** Institute for Advanced Study
- 2018 **CAREER Award** NSF
- 2017 **David and Lucile Packard Foundation Fellowship** David and Lucile Packard Foundation
- 2016 **Outstanding Reviewer** Geophysical Research Letters
- 2015 **Outstanding Reviewer** Geophysical Journal International
- 2012 **AGU Outstanding Student Paper Award** American Geophysical Union
- 2012 **SSA Student Presentation Award** Seismological Society of America
- 2010 **AGU Outstanding Student Paper Award** American Geophysical Union

Teaching

- 2023 - 2025 **Computational/Advanced Seismology**, ESS 563 – U of Washington
- Taught Spring 2023, 2024, 2025
 - Covers ambient noise seismology, earthquake source characterization
 - Julia and Python programming for high-performance computing
- 2023 - 2026 **Introduction to Seismology**, ESS 412/512 – U of Washington
- Taught Winter 2023, 2025, 2026
 - Covers earthquake physics, seismic waves, and hazard assessment
- 2022 - 2026 **Geophysics**, ESS 314 – U of Washington
- Taught 2022, 2023, 2024, 2026
- 2021 - 2024 **Machine Learning in the Geosciences**, ESS 469/569 – U of Washington
- Taught Fall 2021, 2022, 2023, 2024
 - Open-access Jupyter Book with asynchronous teaching materials
 - Course adopted by University of Arizona and UC Berkeley
 - Curated geoscience-specific ML datasets for homework
- 2019 **Machine Learning in Earth and Planetary Sciences**, EPS 268 – Harvard
- Taught Fall 2019
- 2018 **Induced Seismicity**, EPS 268 – Harvard
- Taught Fall 2018
- 2018 - 2020 **Earthquakes and Faulting**, EPS 203 – Harvard
- Taught Fall 2018, 2020

2017 - 2020 **Earthquakes and Tectonics**, EPS 55 – Harvard

- Taught Fall 2017, 2020

2016 **Earthquake Sources**, EPS 204 – Harvard

- Taught Spring 2016

Phd Advisees

2025 - present **Michael Hemmett**, PhD Pre-Candidate – UW ESS

Offshore geophysics

2022 - present **Manuela Koepfli**, PhD Candidate – USS

Geohazard assessment and early warning systems

- 1 publication, 1 in review, 1 in prep

2022 - present **Akash Kharita**, PhD Pre-Candidate – UW ESS

Geohazard characterization using machine learning

- 2 publications

2021 - present **Yiyu Ni**, PhD Candidate – UW ESS

Machine learning and big data seismology

- 8 publications

2019 - 2024 **Congcong Yuan**, PhD – Harvard

Time-dependent seismology, solid-fluid interaction. Next position: Cornell postdoc → Faculty position at NTU Singapore

- 4 publications

2018 - 2023 **Stephanie Olinger**, PhD – Harvard

Cryo-seismology (50% co-advised with Brad Lipovsky). Next position: Stanford Thompson Postdoctoral Fellowship → CEO of AEI

- 4 publications

2016 - 2021 **Tim Clements**, PhD – Harvard

Hydro-seismology and big-data seismology. Next position: USGS Mendenhall Postdoc → USGS Research Geophysicist

- 4 publications

2016 - 2022 **Jiuxun Yin**, PhD – Harvard

Earthquake seismology and source characterization. Next position: Caltech SCSN Postdoc Fellowship → Schlumberger

- 6 publications

Postdocs

2024 - 2025 **Dr. Stephanie Olinger**, University of Washington

Cryoseismology, DAS, Ambient noise monitoring

- Next position: Climate Tech → CEO Applied Environmental Intelligence

2023 - 2024 **Dr. Kuan-Fu Feng**, University of Washington

Environmental Seismology, Cloud computing

- Next position: Postdoc at University of Utah

- 2023 - 2025 **Dr. Ethan Williams**, University of Washington
DAS
• Next position: Assistant Professor at UC Santa Cruz
- 2022 - 2024 **Dr. Qibin Shi**, University of Washington
DAS, Deep Learning, Earthquake Catalog building, AgroSeismology
• Now Postdoctoral Fellow at Rice University
- 2020 - 2022 **Dr. Laura Ermert**, Harvard University/University of Washington
Environmental seismology
• Next position: Assistant Professor at ISTerre
- 2019 - 2020 **Dr. Xiaotao Yang**, Harvard University
Ambient noise cross correlation
• Next position: Assistant Professor at Purdue
- 2019 - 2020 **Dr. Kurama Okubo**, Harvard University
Ambient noise monitoring
• Next position: Researcher at NIED, Japan
- 2019 - 2020 **Dr. Zhitu Ma**, Harvard University
Ambient noise cross correlation
• Next position: Assistant Professor at Tongji University, China
- 2018 - 2019 **Dr. Chengxin Jiang**, Harvard University
Ambient noise seismology, computational seismology
• Next position: Research Associate at Australian National University
- 2016 - 2017 **Dr. Chris Van Houtte**, Harvard University
Earthquake Source seismology
- 2016 - 2018 **Dr. Loïc Viens**, Harvard University
ambient noise seismology, ground motion prediction
• Next position Researcher at Los Alamos

Other Graduate Student Supervision

Note: (*) At UW, roles are secondary advisor or primary advisor on one project/manuscript. (**) Advising resulted in a peer-reviewed publication.

Maleen Kidiwela (*, **): 2021-present, UW Oceanography (co-advised with William Wilcock) - DAS and offshore seismology, resulted in publication

Zoe Krauss (*, **): 2021-2024, UW Oceanography (co-advised with William Wilcock) - Axial Seamount hydrothermal monitoring, resulted in publication

Parker Sprinkle (*, **): 2021-present, UW ESS (co-advised) - Earthquake early warning and machine learning

William Flanagan (*, **): 2019, Harvard EPS - Volcano seismology project

Congcong Yuan (*, **): 2019, USTC China - Master student visiting researcher on time-dependent seismology, resulted in publication

Natasha Toghramadjian (*, **): 2018-2025, Harvard EPS - Seismic attenuation and site effects, resulted in publications

Zhuo Yang (*, **): 2018-2021, Harvard EPS - Machine learning for seismic phase picking, resulted in publication

Philippe Danré (*, **): 2018, Ecole Normale Supérieure Paris - Master thesis on seismic monitoring, resulted in publication

Thibault Pérol (*, **): 2017, Harvard EPS - Deep learning for earthquake detection, resulted in multiple publications

Manuel Florez (*, **): 2017-2019, MIT - Dissertation committee member for geophysics PhD

Graduate Student Committees

Jones, Randall: Atmospheric Sciences, GSR, 2025-present

Koepfli, Manuela: Earth and Space Sciences, Chair, 2024-present

Ni, Yiyu: Earth and Space Sciences, Chair, 2024-present

Pearson, Anna Elaine Rogers: Earth and Space Sciences, Member, 2024-present

Kidiwela, Maleen: School of Oceanography, Member, 2024-present

Sangmin, Song: School of Oceanography, GSR, 2024-present

Sweeney, Aodhan: Atmospheric Sciences, GSR, 2024-present

Kharita, Akash: Earth and Space Sciences, Advisor, 2023-present

DeGrande, Jensen: Earth and Space Sciences, Member, 2023-present

Zhang, Maochuan: School of Oceanography, GSR, 2023-present

Chien, Mu-Ting: Atmospheric Sciences, GSR, 2023-2024

Ragland, John: Electrical and Computer Engineering, GSR, 2023-2024

Velappan, Hemalatha: School of Environmental and Forest Sciences, GSR, 2023-present

Velegar, Meghana S: Applied Mathematics, GSR, 2023-2023

Sprinkle, Parker: Earth and Space Sciences, Member, 2022-present

Krauss, Zoe: School of Oceanography, GSR, 2022-2024

Rasanen, Ryan: Civil and Environmental Engineering, GSR, 2022-2023

Zahn, Olivia: Physics, GSR, 2022-2024

International Phd Advising

Note: Dissertation and defense evaluating committee members

Marius Paul Isken : 2024 (GFZ, Germany)

Luc Illien : 2023 (GFZ, Germany)

Zoe Renate : 2023 (Universite de Lorraine, France)

Reza Esfahani : 2022 (GFZ, Germany)

Kurama Okubo : 2019 (ENS, Paris, France)

Media Coverage

- [2025 Global-scale Database of Seismic Phases](#)
- [2024 Ambient Field Monitoring in the Shallow Crust](#)
- [2024 Making Big Data Look Small](#)
- [2023 UW eScience Seminar](#)
- [2023 Open Science in Seismology](#)
- [2023 UW News Fiber Sensing](#)
- [2018 Radcliffe Seminar: Where we stand in Earthquake Prediction](#)

Undergraduate Students

Note: My undergraduate advising includes research opportunities through summer programs and independent studies (ESS 499 for credit during academic year, hourly pay during summer/breaks). I mentor students toward conference presentations and publications, and support competitive fellowship applications. Legend: (*) Conference presentation | (**) Peer-reviewed publication | (***) Publication in prep | (+) NSF GRFP recipient

Derek Yao: 2026-present, CSE - various agents for Geosciences

Alex Rose : 2025-present, UW Oceanography - DAS data processing research

Anjani Mirchandi: 2024-present, UW Applied Math - DAS data processing research

Hiroto Bito: 2023-2026, UW ESS - ML for earthquake catalog building and spatio-temporal forecasting

Nicholas Wolfe: 2023, UW ESS - Earthquake magnitude estimation project

Informatics Capstone Team (lead Rona Guo*): 2023, Nathan Limono, William Phan, Michael Yung, Matthew Herradura, UW Informatics - DAS Platform development, resulted in conference presentation

Lucas Swanson : 2023, UW Informatics - DAS software development

Francesca Skene (*) : 2022-2023, UW ESS - Surface event catalog, resulted in conference presentation

Nick Smoczyk (*, **) : 2022-2024, University of Minnesota - Volcano seismology using RedPy catalog, resulted in conference presentation and publication in prep

Julian Schmitt (*, +) : 2020-2021, Harvard EPS - Ambient noise seismology on cloud computing with Julia, BASIN project, resulted in conference presentation and NSF GRFP

Jared Bryan (*, **, +): 2019, Harvard EPS - Ambient noise monitoring, SCEC Internship, resulted in conference presentation, publication, and NSF GRFP

Albert Aguilar (*): 2018, Harvard EPS - Subduction zone seismology and data mining, SCEC, resulted in conference presentation

Leore Lavin: 2016, Harvard EPS - Ground motion simulations with ambient noise

Roy Bowling (*): 2014, Scripps IGPP - Ambient noise seismology, SCEC Internship

Tara Larrue (*): 2012, Stanford Geophysics - Ambient noise seismology, SURGE program

Penprapa Wutthijuk (*): 2011, Stanford - Ambient noise seismology, SURGE program

Training Workshops

- 2025 [CRESCENT-SCOPED Workshop](#) co-PI organizer for ML and Earthquake Catalog Building workshop (35 participants)
- 2024 [SCOPED Workshop](#) Lead PI organizer for Cloud/HPC/wavefield simulations workshop (100 participants)
- 2024 [SSA Cloud 101 & Data Mining](#) Lead organizer of cloud workshop (80 participants)
- 2023 [CyberTraining Workshop](#) Lead PI, coordinator, and instructor for HPC and Data Science
- 2021 – present [GeoSMART ML for Solid Earth Science](#) Co-organizer of ML training with Jupyter Book
- 2016 **SCEC-ERI VISES Summer School** Scientific committee member and instructor

University Service

- 2026 – present **AI Minor task force member**, University of Washington
- reviewing AI Minor proposal for cross-university program, ~ 1hr/week for spring quarter
- 2026 – present **Department Colloquium Committee**, University of Washington
- Help organizing the department colloquium series
- 2026 – present **AI@UW Advisory Committee to the VP**, University of Washington
- Advising AI strategy and initiatives across the university, commitment ~ 1hr/week
- 2024 – present **Co-director of UW CS4Env**, University of Washington
- Computer Science for the Environment (CS4Env) is a cross-unit initiative to promote synergies between computer science and environmental research at UW, includes organizing biweekly seminar, reviewing proposals, and organizing an annual symposium.
- 2024 – present **Chair of Curriculum Committee**, UW/ESS
- Reviewing proposed courses, course changes, undergraduate and graduate program.
- 2022 – present **Member of Curriculum Committee and Data Science Oversight Committee**, UW/ESS
- Review the proposed data science option program
- 2022 – 2023 **Member of Executive Committee**, UW/ESS
- 2022 – 2023 **Member of Research Faculty Search Committees**, UW/ESS
- Search committee for PNSN Network Manager and Igneous Processes Faculty Search
- 2016 – 2020 **Multiple Committee Memberships**, Harvard University
- Undergraduate Curriculum Committee, Graduate Student Council, IT Committee, Diversity Inclusion and Belonging Committee, Department Colloquium Committee

Professional Service

- 2026 – present **Member of Board of Directors at Earthscope Consortium**,
- 2024 – 2025 **Member of Science Steering Committee at SCEC**,
- 2023 – 2025 **Chair of Integration and Innovation Advisory Committee at Earthscope Consortium**,
- 2022 – 2022 **Committee Member of Charles Richter Early Career Award Committee at Seismological Society of America**,
- 2021 – 2022 **Member of Data Service Standing Committee at IRIS**,

- 2021 – 2023 **Member of HPC Standing Committee at SCEC,**
2011 – 2012 **Chair of Graduate Student Council at Stanford University,**

Review Panels

- 2020,2024 **NSF Review Panels EAR-Geophysics and GeoInformatics,**
• Reviewed 10+ proposals per panel
2016 – 2018 **USGS Review Panels Earthquake Hazards Program,**
• Reviewed 10+ proposals per panel

Editorial Service

- 2025 – present **Editor at Geophysical Journal International,**
2017 – 2020 **Associate Editor at Geophysical Research Letters,**
2014 – present **Reviewer for multiple journals in geophysics and seismology,**
• >150 reviews for GJI, BSSA, Nature Communications, GRL, JGR, Science, and others

Publications Legend

Author Notation: Bold: Marine A. Denolle (PI). *Italic:* graduate students, postdocs, and undergraduate students mentored by MD.

Publications

- 2026 **Distributed Acoustic Sensing Records of Earthquakes and Surface Processes at Mount Rainier Volcano**
Veronica Gaete-Elgueta, *Manuela Köpfli*, Dominik Gräff, Bradley P Lipovsky, **Marine A. Denolle**, Weston A. Thelen, Brendan Pratt, Taylor R. Kenyon
(in review in Seismica)
- 2026 **Agroseismology and the impact of farming practices on soil hydrodynamics**
Qibin Shi, David R. Montgomery, Abigail L.S. Swann, Nicoleta C. Cristea, *Ethan F. Williams*, Nan You, Simon Jeffery, Joe Collins, Ana Prada Barrio, Paula A. Misiewicz, Tarje Nissen-Meyer, **Marine A. Denolle**
[10.1126/science.aec0970](https://doi.org/10.1126/science.aec0970) (Science, 392)
- 2026 **Active prot thrusts and fluid highways: Seismic noise reveals hidden subduction dynamics in Cascadia**
Maleen Kidiwela, **Marine A. Denolle**, William S. D. Wilcock, *Kuan-Fu Feng*
[10.1126/sciadv.aea3684](https://doi.org/10.1126/sciadv.aea3684) (Science Advances, 12)
- 2026 **DeepSubDAS: an earthquake phase picker from submarine distributed acoustic sensing data**
Han Xiao, Frederik Tilmann, Martijn van den Ende, Diane Rivet, Afonso Loureiro, Takeshi Tsuji, Arantza Ugalde, *Qibin Shi*, **Marine A Denolle**
[10.1093/gji/ggag061](https://doi.org/10.1093/gji/ggag061) (Geophysical Journal International, 245)
- 2026 **Exploration of Machine Learning Methods to Seismic Event Discrimination in the Pacific Northwest**
Akash Kharita, **Marine Denolle**, Alexander Hutko, Renate Hartog, Stephen Malone
[10.26443/seismica.v5i1.2068](https://doi.org/10.26443/seismica.v5i1.2068) (Seismica, 5)
- 2026 **Seismological Analysis of Contemporary and Future Alpine Fault Earthquakes Using the Southern Alps Long Skinny Array (SALSA)**
John Townend, Ilma del Carmen Juarez-Garfias, Olivia Pita-Sllim, Calum J. Chamberlain, Emily Warren-Smith, Caroline Holden, Kasper van Wijk, **Marine Denolle**, Andrew Curtis, Hiroe Miyake
[10.1785/0220250260](https://doi.org/10.1785/0220250260) (Seismological Research Letters)

- 2026 **Probing the Seattle Basin Edge Using a Dense Urban Nodal Array in 100 Backyards**
Natasha Toghramadjian, Marine A. Denolle, Laura Ermert, Chengxin Jiang
[10.1785/0220250241](https://doi.org/10.1785/0220250241) (Seismological Research Letters)
- 2026 **The Seismic Wavefield Common Task Framework**
Alexey Yermakov, Yue Zhao, Marine Denolle, Yiyu Ni, Philippe M. Wyder, Judah Goldfeder, Stefano Riva, Jan Williams, David Zoro, Amy Sara Rude, Matteo Tomasetto, Joe Germany, Joseph Bakarji, Georg Maierhofer, Miles Cranmer, J. Nathan Kutz
[10.48550/arXiv.2512.19927](https://arxiv.org/abs/10.48550/arXiv.2512.19927) (arXiv)
- 2026 **A Decadal Survey of the Near-Surface Seismic Velocity Response to Hydrological Variations in Utah, United States**
Kuan-Fu Feng, Marine Denolle, Fan-Chi Lin, Tonie van Dam
 (Journal of Geophysical Research: Solid Earth, 131)
- 2025 **A Global-scale Database of Seismic Phases from Cloud-based Picking at Petabyte Scale**
Yiyu Ni, Marine Denolle, Amanda Thomas, Alex Hamilton, Jannes Münchmeyer, Yinzhi Wang, Loïc Bachelot, Chad Trabant, David Mencin
[10.26443/seismica.v4i2.1738](https://doi.org/10.26443/seismica.v4i2.1738) (Seismica, 4)
- 2025 **A Review of Cloud Computing and Storage in Seismology**
Yiyu Ni, Marine A Denolle, Jannes Münchmeyer, Yinzhi Wang, Kuan-Fu Feng, Carlos Garcia Jurado Suarez, Amanda M Thomas, Chad Trabant, Alex Hamilton, David Mencin
[10.1093/gji/ggaf322](https://doi.org/10.1093/gji/ggaf322) (Geophysical Journal International)
- 2025 **Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing Through Diverse Training Modalities**
Marine A. Denolle, Carl Tape, Ebru Bozdağ, Yinzhi Wang, Felix Waldhauser, Alice-Agnes Gabriel, Jochen Braunmiller, Bryant Chow, Liang Ding, Kuan-Fu Feng, Ayon Ghosh, Nathan Groebner, Aakash Gupta, Zoe Krauss, Amanda M. McPherson, Masaru Nagaso, Zihua Niu, Yiyu Ni, Rıdvan Örsveran, Gary Pavlis, Felix Rodriguez-Cardozo, Theresa Sawi, David Schaff, Nico Schliwa, David Schneller, Qibin Shi, Julien Thurin, Chenxiao Wang, Kaiwen Wang, Jeremy Wing Ching Wong, Sebastian Wolf, Congcong Yuan
[10.1785/0220240413](https://doi.org/10.1785/0220240413) (Seismological Research Letters, 96)
- 2025 **Ambient field seismology in critical zone hydrological sciences**
Marine A. Denolle, Qibin Shi, Tim Clements, Loïc Viens, Veronica Rodriguez-Tribaldos, Fabrice Cotton
[10.5802/crgeos.310](https://doi.org/10.5802/crgeos.310) (Comptes Rendus. Géoscience, 357)
- 2025 **Denoising Offshore Distributed Acoustic Sensing Using Masked Auto-Encoders to Enhance Earthquake Detection**
Q. Shi, M. A. Denolle, Y. Ni, E. F. Williams, N. You
[10.1029/2024JB029728](https://doi.org/10.1029/2024JB029728) (JGR: Solid Earth, 130)
- 2025 **Multiplexed Distributed Acoustic Sensing Offshore Central Oregon**
Q. Shi, E. F. Williams, B. P. Lipovsky, M. A. Denolle, W. S. D. Wilcock, D. S. Kelley, K. Schoedl
[10.1785/0220240460](https://doi.org/10.1785/0220240460) (Seismological Research Letters, 96)
- 2024 **Wavefield reconstruction of distributed acoustic sensing: Lossy compression, wavefield separation, and edge computing**
Y. Ni, M. A. Denolle, Q. Shi, B. P. Lipovsky, S. Pan, J. N. Kutz
[10.1029/2024JH000247](https://doi.org/10.1029/2024JH000247) (Journal of Geophysical Research: Machine Learning and Computation, 1)
- 2024 **Analysing Volcanic, Tectonic, and Environmental Influences on the Seismic Velocity from 25 Years of Data at Mount St. Helens**
P. Makus, M. A. Denolle, C. Sens-Schönfelder, M. Köpfl, F. Tilmann
[10.1785/0220240088](https://doi.org/10.1785/0220240088) (Seismological Research Letters, 95)
- 2024 **Examining 22 Years of Ambient Seismic Wavefield at Mount St. Helens**
M. Köpfl, M. A. Denolle, W. Thelen, P. Makus, S. Malone
[10.1785/0220240079](https://doi.org/10.1785/0220240079) (Seismological Research Letters, 95)

- 2024 **Impact of Temperature and Relative Humidity Variations on Coda Waves in Concrete**
F. Diwald, **M. Denolle**, J. J. Timothy, C. Gehlen
[10.1038/s41598-024-69564-4](https://doi.org/10.1038/s41598-024-69564-4) (Scientific Reports, 14)
- 2024 **Monitoring velocity change over 20 years at Parkfield**
K. Okubo, B. Delbridge, **M. Denolle**
[10.1029/2023JB028084](https://doi.org/10.1029/2023JB028084) (Journal of Geophysical Research: Solid Earth, 129)
- 2024 **Extended crack propagation by local nucleation and rapid transverse expansion**
T. Cochard, I. Svetlizky, G. Albertini, R. C. Viesca, S. M. Rubinstein, F. Spaepen, C. Yuan, **M. Denolle**, Y.-Q. Song, L. Xiao, D. A. Weitz
[10.1038/s41567-023-02365-0](https://doi.org/10.1038/s41567-023-02365-0) (Nature Physics)
- 2023 **Discrimination between icequakes and earthquakes in southern Alaska: an exploration of waveform features using random forest algorithm**
A. Kharita, **M. Denolle**, M. West
[10.1093/gji/ggae106](https://doi.org/10.1093/gji/ggae106) (Geophysical Journal International)
- 2023 **Ocean Coupling Limits Rupture Velocity of Fastest Observed Ice Shelf Rift Propagation Event**
S. Olinger, B. Lipovsky, **M. Denolle**
[10.1029/2023AV001023](https://doi.org/10.1029/2023AV001023) (AGU Advances, 5)
- 2023 **Laboratory hydrofracture as analogs to tectonic tremors**
C. Yuan, T. Cochard, **M. Denolle**, J. Gombert, A. Wech, L. Xiao, D. Weitz
[10.1029/2023AV001002](https://doi.org/10.1029/2023AV001002) (AGU Advances, 5)
- 2023 **Improved observations of deep earthquake ruptures using machine learning**
Q. Shi, **M. Denolle**
[10.1029/2023JB027334](https://doi.org/10.1029/2023JB027334) (Journal of Geophysical Research: Solid Earth, 128)
- 2023 **Better Together: Ensemble Learning for Earthquake Detection and Phase Picking**
Congcong Yuan, Yiyu Ni, Youzuo Lin, **Marine Denolle**
[10.1109/TGRS.2023.3320148](https://doi.org/10.1109/TGRS.2023.3320148) (IEEE Transactions on Geoscience and Remote Sensing, 61)
- 2023 **An Object Storage for Distributed Acoustic Sensing**
Yiyu Ni, **Marine A. Denolle**, Rob Fatland, Naomi Alterman, Bradley P. Lipovsky, Friedrich Knuth
[10.1785/0220230172](https://doi.org/10.1785/0220230172) (Seismological Research Letters, 95)
- 2023 **Seismology in the cloud: guidance for the individual researcher**
Z. Krauss, Y. Ni, S. Henderson, **M. Denolle**
[10.26443/seismica.v2i2.979](https://doi.org/10.26443/seismica.v2i2.979) (Seismica, 2)
- 2023 **Curated Pacific Northwest AI-ready Seismic Dataset**
Y. Ni, A. Hutko, F. Skene, **M. Denolle**, S. Malone, P. Bodin, R. Hartog, A. Wright
[10.26443/seismica.v2i1.368](https://doi.org/10.26443/seismica.v2i1.368) (Seismica, 2)
- 2023 **Probing environmental and tectonic changes underneath Ciudad de México with the urban seismic field**
L. Ermert, E. Cabral-Cano, E. Chaussard, D. Solano-Rojas, L. Quintanar, D. Morales Padilla, E. A. Fernandez-Torres, **M. A. Denolle**
[10.5194/egusphere-2022-1361](https://doi.org/10.5194/egusphere-2022-1361) (Solid Earth (EGU))
- 2023 **The Seismic Signature of California's Earthquakes, Droughts, and Floods**
T. Clements, **M. A. Denolle**
[10.1029/2022JB025553](https://doi.org/10.1029/2022JB025553) (Journal of Geophysical Research: Solid Earth, 128)
- 2022 **A multitask encoder-decoder to separate earthquake and ambient noise signal in seismograms**
J. Yin, **M. A. Denolle**, B. He
[10.1093/gji/ggac290](https://doi.org/10.1093/gji/ggac290) (Geophysical Journal International, 231)

- 2022 **Pronounced Seismic Anisotropy in Kanto Sedimentary Basin: A Case Study of Using Dense Arrays, Ambient Noise Seismology, and Multi-Modal Surface-Wave Imaging**
C. Jiang, M. A. Denolle
[10.1029/2022JB024613](https://doi.org/10.1029/2022JB024613) (Journal of Geophysical Research: Solid Earth, 127)
- 2022 **Imaging the Kanto Basin seismic basement with earthquake and noise autocorrelation functions**
L. Viens, C. Jiang, M. A. Denolle
[10.1093/gji/ggac101](https://doi.org/10.1093/gji/ggac101) (Geophysical Journal International, 230)
- 2022 **Tracking the Cracking: a Holistic Analysis of Rapid Ice Shelf Fracture Using Seismology, Geodesy, and Satellite Imagery on the Pine Island Glacier Ice Shelf, West Antarctica**
S. D. Olinger, B. P. Lipovsky, M. A. Denolle, B. W. Crowell
[10.1029/2021GL097604](https://doi.org/10.1029/2021GL097604) (Geophysical Research Letters)
- 2022 **Detecting Elevated Pore Pressure due to Wastewater Injection Using Ambient Noise Monitoring**
Z. Yang, C. Yuan, M. A. Denolle
[10.1785/0320210036](https://doi.org/10.1785/0320210036) (The Seismic Record, 2)
- 2021 **The Earth's Surface Controls the Depth-Dependent Seismic Radiation of Megathrust Earthquakes**
J. Yin, M. A. Denolle
[10.1029/2021AV000413](https://doi.org/10.1029/2021AV000413) (AGU Advances, 2)
- 2021 **Comparing approaches to measuring seismic phase variations in the time, frequency, and wavelet domains**
C. Yuan, J. Bryan, M. A. Denolle
[10.1093/gji/ggab140](https://doi.org/10.1093/gji/ggab140) (Geophysical Journal International, 226)
- 2021 **Source time function clustering reveals patterns in earthquake dynamics**
J. Yin, Z. Li, M. A. Denolle
[10.1785/0220200403](https://doi.org/10.1785/0220200403) (Seismological Research Letters, 92)
- 2021 **SeisNoise.jl: Ambient Seismic Noise Cross Correlation on the CPU and GPU in Julia**
T. Clements, M. A. Denolle
[10.1785/0220200192](https://doi.org/10.1785/0220200192) (Seismological Research Letters, 92)
- 2020 **Quiet Anthropocene, quiet Earth**
M. A. Denolle, T. Nissen-Meyer
[10.1126/science.abd8358](https://doi.org/10.1126/science.abd8358) (Science, 369)
- 2020 **SeisIO: A Fast, Efficient Geophysical Data Architecture for the Julia Language**
J. P. Jones, K. Okubo, T. Clements, M. A. Denolle
[10.1785/0220190295](https://doi.org/10.1785/0220190295) (Seismological Research Letters, 91)
- 2020 **NoisePy: A new high-performance python tool for ambient-noise seismology**
C. Jiang, M. A. Denolle
[10.1785/0220190364](https://doi.org/10.1785/0220190364) (Seismological Research Letters, 91)
- 2019 **Earthquakes Within Earthquakes: Patterns in Rupture Complexity**
P. Danré, J. Yin, B. Lipovsky, M. Denolle
[10.1029/2019GL083093](https://doi.org/10.1029/2019GL083093) (Geophysical Research Letters, 43)
- 2019 **Long-period ground motions from past and virtual megathrust earthquakes along the Nankai Trough, Japan**
L. Viens, M. Denolle
[10.1785/0120180320](https://doi.org/10.1785/0120180320) (Bulletin of the Seismological Society of America, 109)
- 2019 **Energetic Onset of Earthquakes**
M. Denolle
[10.1029/2018GL080687](https://doi.org/10.1029/2018GL080687) (Geophysical Research Letters, 46)

- 2019 **Relating teleseismic backprojection images to earthquake kinematics**
J. Yin, M. Denolle
[10.1093/gji/ggz048](https://doi.org/10.1093/gji/ggz048) (Geophysical Journal International, 217)
- 2019 **Geometric Controls on Pulse-like Rupture in a Dynamic Model of the 2015 Gorkha Earthquake**
Y. Wang, M. Denolle, S. M. Day
[10.1029/2018JB016602](https://doi.org/10.1029/2018JB016602) (Journal of Geophysical Research, 124)
- 2018 **Tracking ground water using the ambient seismic field**
T. Clements, M. Denolle
[10.1029/2018GL077706](https://doi.org/10.1029/2018GL077706) ((User text suggests possible mismatch of volume/issue) Geophysical Research Letters, 123)
- 2018 **Complex near-surface rheology inferred from the response of greater Tokyo to strong ground motions**
L. Viens, M. Denolle, N. Hirata, S. Nakagawa
 (Journal of Geophysical Research: Solid Earth, 123)
- 2018 **Strong Shaking Predicted in Tokyo From an Expected M7+ Itoigawa-Shizuoka Earthquake**
M. A. Denolle, P. Boué, N. Hirata, G. C. Beroza
 (Journal of Geophysical Research: Solid Earth, 123)
- 2018 **Improved model fitting for the empirical Green's function approach using hierarchical models**
C. Van Houtte, M. Denolle
 (Journal of Geophysical Research: Solid Earth, 123)
- 2018 **Tracking groundwater levels using the ambient seismic field**
T. Clements, M. A. Denolle
[10.1029/2018GL077706](https://doi.org/10.1029/2018GL077706) (Geophysical Research Letters, 45)
- 2018 **Convolutional neural network for earthquake detection and location**
T. Perol, M. Gharbi, M. Denolle
 (Science Advances, 4)
- 2018 **Spatial and Temporal Evolution of Earthquake Dynamics: Case Study of the Mw 8.3 Illapel Earthquake, Chile**
J. Yin, M. A. Denolle, H. Yao
[10.1002/2017JB014265](https://doi.org/10.1002/2017JB014265) (Journal of Geophysical Research: Solid Earth, 123)
- 2017 **Multicomponent C3 Green's Functions for Improved Long-Period Ground-Motion Prediction**
Y. Sheng, M. A. Denolle, G. C. Beroza
[10.1785/0120170053](https://doi.org/10.1785/0120170053) (Bulletin of the Seismological Society of America, 107)
- 2017 **Retrieving impulse response function amplitudes from the ambient seismic field**
L. Viens, M. Denolle, H. Miyake, S. Sakai, S. Nakagawa
[10.1093/gji/ggx155](https://doi.org/10.1093/gji/ggx155) (Geophysical Journal International, 210)
- 2016 **Beyond Basin Resonance: Characterizing Wave Propagation Using a Dense Array and the Ambient Seismic Field**
P. Boue, M. Denolle, N. Hirata, S. Nakagawa, G. C. Beroza
[10.1093/gji/ggw205](https://doi.org/10.1093/gji/ggw205) (Geophysical Journal International, 206)
- 2016 **New perspective on self-similarity of shallow thrust earthquakes**
M. Denolle, P. M. Shearer
[10.1002/2016JB013105](https://doi.org/10.1002/2016JB013105) (Journal of Geophysical Research: Solid Earth, 121)
- 2015 **Dynamics of the M7.8 2015 Nepal Earthquake**
M. Denolle, W. Fan, P. M. Shearer
[10.1002/2015GL065336](https://doi.org/10.1002/2015GL065336) (Geophysical Research Letters, 42)

- 2014 **Full 3D Tomography (F3DT) for Crustal Structure in Southern California Based on the Scattering-Integral (SI) and the Adjoint-Wavefield (AW) Methods**
E.-J. Lee, P. Chen, T. H. Jordan, P. B. Maechling, **M. Denolle**, G. C. Beroza
[10.1002/2014JB011236](https://doi.org/10.1002/2014JB011236) (Journal of Geophysical Research, 119)
- 2014 **Long-period seismic amplification in the Kanto Basin from the ambient seismic field**
M. Denolle, H. Miyake, S. Nakagawa, N. Hirata, G. C. Beroza
[10.1002/2014GL059425](https://doi.org/10.1002/2014GL059425) (Geophysical Research Letters, 41)
- 2014 **Strong Ground Motion Prediction using Virtual Earthquakes**
M. Denolle, E. M. Dunham, G. A. Prieto, G. C. Beroza
[10.1126/science.1245678](https://doi.org/10.1126/science.1245678) (Science, 343)
- 2013 **Ground Motion Prediction of Realistic Earthquake Sources Using the Ambient Seismic Field**
M. Denolle, E. M. Dunham, G. A. Prieto, G. C. Beroza
[10.1029/2012JB009603](https://doi.org/10.1029/2012JB009603) (Journal of Geophysical Research, 118)
- 2013 **A numeric evaluation of attenuation from ambient noise correlation functions**
J. F. Lawrence, **M. Denolle**, K. J. Seats, G. Prieto
[10.1002/2012JB009513](https://doi.org/10.1002/2012JB009513) (Journal of Geophysical Research, 118)
- 2012 **Solving the Surface-Wave Eigenproblem with Chebyshev Spectral Collocation**
M. Denolle, E. M. Dunham, G. C. Beroza
[10.1785/0120110183](https://doi.org/10.1785/0120110183) (Bulletin of the Seismological Society of America, 102)
- 2011 **On amplitude carried by the ambient seismic field**
G. A. Prieto, **M. Denolle**, J. F. Lawrence, G. C. Beroza
(Comptes Rendus Geoscience (Thematic Issue: Imaging and Monitoring with Seismic Noise), 343)
- 2010 **Evidence of active backthrusting at the NE Margin of Mentawai Islands, SW, Sumatra**
S. Singh, N. Hananto, A. Chauhan, H. Permana, **M. Denolle**, A. Hendriyana, D. Natawidjaja
[10.1111/j.1365-246X.2009.04458.x](https://doi.org/10.1111/j.1365-246X.2009.04458.x) (Geophysical Journal International, 180)

Grants

- 2025 **Jerry and Linda Paros** Multi-Sensor, Multi-Geohazards Monitoring at Mt Rainier and Mountaineous regions: \$750,000 (lead PI)
- 2025 **UW CRESST - FFST** Toward Predictive Understanding of Climate-Compounded Geohazards: \$300,000 (lead PI)
- 2025 **NSF CS4AII** Separating the Signal from the Noise: Promoting Alaskan students inquiry with geographically relevant seismic data and machine learning techniques: \$889,766 (lead PI Lore, Denolle co-PI, \$62,000 to UW), DRL-2524060
- 2025 **NSF RISE CAIG** Collaborative Research: CAIG: Framework for Artificial Intelligence-Enhanced Modeling of Wildfire Geohazards (FAIM-WG): Applications for postfire debris flows across the Western US: \$735,074 (lead PI Istanbuluoglu, Denolle co-PI), RISE-2530591
- 2025 **NSF OCE** Multi-span distributed fiber sensing on the Ocean Observatories Initiative Regional Cabled Array: \$880,490 (lead PI Wilcock, Denolle co-PI), OCE-2526198, technology development grant.
- 2025 **NSF OCE EXTension** on the ENDeavour Segment (EXTEND): Illuminating the seafloor spreading cycle: \$785,819.00 (lead PI Wilcock, Denolle co-PI), OCE-2439442
- 2024 **NSF EAR** Collaborative Research: A seismic investigation of slow slip and fault locking along the Alaska-Aleutian subduction zone: \$248,835.00 (Denolle co-PI, collaborative with lead PI Golos), EAR-2346079

- 2023 **NSF OCE RAPID**: Multiplexed Distributed Acoustic Sensing (DAS) at the Ocean Observatory Initiative (OOI) Regional Cabled Array (RCA): \$198,069.00 (lead PI Wilcock, Denolle co-PI), OCE-2415521
- 2023 **Ecole Normal Supérieure de Paris** Travel Fellowship: \$3,500 (Denolle PI)
- 2023 **SCEC CyberTraining** for Seismology: Data Science and HPC: \$35,229 (Denolle PI)
- 2021 **David and Lucile Packard Foundation** URG Program: \$50,000 (PI)
- 2021 **Murdock Charitable Trust Fund** UW FiberLab equipment: \$950,000 (co-PI, lead PI Brad Lipovsky). Built facility for Fiber Sensing.
- 2021 **NSF CyberTraining** GeoSMART ML workforce development: \$995,817 (co-PI). Build a course MLGEO.
- 2021 **NSF OAC-CSSI** Collaborative Research: Frameworks: Seismic COmputational Platform for Empowering Discovery (SCOPED): \$717,441.00 (Denolle PI, lead-PI Carl Tape), OAC-2103701
- 2021 **NSF EAR, PREEVENTS** Collaborative Research: Cross-Validation of Empirical and Physics-based ground motion predictions: \$167,804.00 (Denolle PI), EAR-2125337
- 2020 **SCEC grant** Aftershock patterns and co-seismic off-fault damage elucidate dynamic rupture processes on the 2019 Ridgecrest earthquake sequence: \$33,307 (Denolle PI) - returned
- 2019 **Harvard University - David Rockefeller Center for Latin American Studies** Monitoring Seismic Hazards in Mexico City using Grillo, a Low-Cost Earthquake Early Warning System, Hardware, \$85,000 (Denolle PI)
- 2019 **Harvard Data Science Initiative** Ambient-noise seismology using Cloud Computing: \$27,210 (Denolle PI)
- 2018 **NSF EAR, CAREER** CAREER: Dynamics of surface rupturing thrust earthquakes: \$504,315 (PI)
- 2018 **SCEC grant** Data Collection for Virtual Earthquakes on Cajon Pass: \$28,085 (Denolle PI) - field work
- 2017 **David and Lucile Packard Foundation** Fellowship: \$875,000 (PI)
- 2017 **NSF PREEVENTS** Collaborative Proposal - PREEVENTS Track 2: Cascadia Scenario Earthquakes: Source, Path, and implications for Earthquake Early Warning: \$324,495.00 (Denolle PI, collaborative with lead PI Yihe Huang and Amanda Thomas), EAR-1739712
- 2017 **SCEC grant** Static and dynamic source parameters of global strike-slip earthquakes: \$25,000 (Denolle PI)
- 2016 **SCEC grant** Epistemic uncertainties in ground motion prediction from virtual earthquakes: \$26,173 (Denolle PI)
- 2015 **SCEC grant** Basin Response to Virtual Earthquakes on the San Jacinto Fault and the Itoigawa-Shizuoka Fault: \$20,000 (Denolle PI)

Research Products

- 2025 **Global Seismic Phase Database** Dataset
Global seismic data product from cloud-scale waveform processing. [DOI](#).
- Scale: 4.3 billion phase picks from 1.3 PB of continuous seismic data.
 - Publication link: Seismica (2025) with DOI-backed archival record.
 - Supports global monitoring and data-driven geophysics workflows.
- 2023 **EarthML4PNW / PNW AI-Ready Seismic Dataset** Dataset
Curated ML-ready Pacific Northwest dataset product. [Dataset repository](#).
- Publication link: Seismica (2023), DOI [10.26443/seismica.v2i1.368](#).
 - Version-controlled release and metadata stewardship through GitHub workflows.
 - Designed for reproducible benchmarking and graduate-level ML coursework.
- 2021 – present **Machine Learning in the Geosciences (Open-Access Jupyter Book)** Textbook
[Book](#) and [course repository](#) provide graduate-level ML curriculum, asynchronous modules, and home-work data pipelines for geoscience training.
- Addresses a core gap: no canonical ML textbook existed for geoscience graduate classrooms.
 - Adoption and contribution interest from peer programs (including Arizona and Berkeley).
 - Ongoing community development with curated datasets and reproducible teaching workflows.
- 2020 – present **SeisNoise.jl** Software
Open-source Julia software for high-performance ambient-noise processing and cross-correlation. [GitHub](#).
- Extends the Julia seismology ecosystem with core ambient-noise tooling.
 - Community signal: 50 stars and 17 forks (Apr 2023 snapshot).
 - Used by group members for production research workflows.
- 2019 – present **NoisePy** Software
Open-source Python software for large-scale ambient-noise seismology. [GitHub](#).
- Community scale: 207 stars, 83 forks, 19 contributors (Jan 2026).
 - Operational usage: taught in workshops and used for large-scale cross-correlation workflows.
 - Open, version-controlled development supporting reproducible and cloud-oriented processing.

Invited Talks

- 2026 **National Academy of Sciences - COSEG** Seminar Speaker
- 2026 **NGF Science Meeting** Conference Speaker
- 2026 **WCCM 2026, Munich** Keynote Speaker
- 2026 **European Geophysical Union 3x** Conference Speaker
- 2026 **CERI Memphis** Department Colloquium
- 2025 **IEEE New Era AI World Leader Summit, Bothell** Keynote Speaker
- 2025 **SIAM Geoscience Meeting** Plenary Speaker
- 2025 **Workshop on Earthquake Physics and Applications of AI to Tectonic Faulting, Italy** Plenary Speaker
- 2025 **Civil Environmental Engineering, University of Washington** Department Colloquium

2025 **Radcliffe Institute of Advanced Studies, on the road** Short Talk

2025 **Washington University, Saint Louis** Department Colloquium

2024 **University of Southern California** Department Colloquium

2024 **University of California, Davis** Department Colloquium

2024 **Northern Arizona University** Department Colloquium

2024 **Passive imaging and monitoring in wave physics (Cargese, France)** Invited Conference Talk

2024 **Statewide California Earthquake Center** Plenary Speaker

2023 **Ecole Normale Supérieure, Paris Séminaire** Departemental

2023 **eScience Institute, University of Washington** Data Science Seminar

2023 **Sandia National Lab, GNEM seminar series** Department Colloquium

2022 **American Geophysical Union (x2)** Invited Conference Talk

2022 **University of New Mexico** Department Colloquium

2021 **University of Oregon** Seismo Colloquium

2021 **U Utah, Seismo Tea** Seismo Colloquium

2021 **University of Wisconsin** Department Colloquium

2021 **Colorado School of Mines** Department Colloquium

2021 **Mexico a traves de los sismos** Invited Conference Talk

2020 **University of Washington** Department Colloquium

2020 **UC Berkeley** Department Colloquium

2019 **Yale University** Department Colloquium

2019 **University of Washington, seismolunch** Seismo Colloquium

2019 **EGU annual meeting, Vienna** Invited Conference Talk

2019 **Michigan State University** Department Colloquium

2019 **Victoria University of Wellington, SN Jepson Lecture, New Zealand** Public Lecture

2019 **GNS-Science, New Zealand** Department Colloquium

2019 **Stanford University, Department of Geophysics** Department Colloquium

2019 **Tufts University, Civil Engineering seminar** Department Colloquium

2018 **Brown University** Department Colloquium

- 2018 **Ecole Normale Supérieure, Paris** Department Colloquium
- 2017 **AGU, New Orleans** Invited Conference Talk
- 2017 **Columbia University, Lamont Doherty Earth Observatory** Department Colloquium
- 2016 **Harvard Museum of Natural History** Public Lecture
- 2016 **University of Oregon** Department Colloquium
- 2016 **University of New Hampshire, Chapman Colloquium** Department Colloquium
- 2016 **UC Santa Cruz, IGPP seminar** Department Colloquium
- 2016 **Massachusetts Institute of Technology** Department Colloquium
- 2015 **USGS, Menlo Park, Earthquake Hazard Program** Department Colloquium
- 2015 **University of Victoria, BC, Canada** Department Colloquium
- 2015 **Penn State, Geodynamics seminar** Department Colloquium
- 2015 **Harvard, Earth and Planetary Sciences** Department Colloquium
- 2015 **UT Austin, Solid Earth seminar** Department Colloquium
- 2015 **UCLA, seismology/tectonics seminar** Department Colloquium
- 2015 **HOKUDAN International Symposium on Active Faulting, Awaji, Japan** Invited Conference Talk
- 2015 **Information Theory and Applications workshop, La Jolla** Invited Conference Talk
- 2015 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium
- 2015 **University of Southern California, Geophysics seminar** Department Colloquium
- 2014 **Strong Motion Studies for Future Mega-Quakes, DPRI, Kyoto University, Japan** Invited Conference Talk
- 2014 **AGU-SEG Summer Research workshop, Vancouver, Canada** Invited Conference Talk
- 2014 **San Diego State University** Department Colloquium
- 2014 **UC Santa Barbara** Department Colloquium
- 2014 **IGPP-Scripps Institution of Oceanography, UCSD, Geophysics seminar** Department Colloquium
- 2013 **AGU, Meeting of the Americas, Cancun, Mexico** Invited Conference Talk
- 2013 **Caltech Seismo Lab** Department Colloquium
- 2013 **USGS, Menlo Park** Department Colloquium
- 2013 **Stanford ICME seminar** Department Colloquium
- 2013 **Earthquake Research Institute, Tokyo University, Japan** Department Colloquium

- 2013 **Advanced Industrial Science and Technology, Japan** Department Colloquium
- 2013 **Disaster Prevention Research Institute, Japan** Department Colloquium
- 2012 **ACOUSTICS, France** Invited Conference Talk
- 2012 **Berkeley Seismo Lab** Department Colloquium
- 2011 **Institut de Physique du Globe de Paris, Earthquake seminar** Department Colloquium